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REGULAR ARTICLE

Modelling root system development for anchorage of forest trees up to the mature stage, including acclimation to soil constraints: the case of *Pinus pinaster*

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Abstract

Aims A reliable root architectural model is required to study tree anchorage, including secondary root growth, from seed to the mature stage.

Methods We calibrated the generic RootTyp model for the seven root types of *Pinus pinaster*. We determined most parameter distributions from the means and standard deviations of corresponding features in 92 excavated root systems of various ages. We set the remaining parameters using the literature or an optimisation method. The simulated root systems were compared to 22 additional root systems for validation.

Results Branching varies as a function of the distance from the root base, and the growth capacity decreases with branching order. These damping properties had to be implemented in the model to yield realistic outputs beyond the juvenile phase. Diameter growth was satisfactorily modelled from the apex diameter and a tapering-by-branching coefficient. Nevertheless, the diameters of the simulated roots were underestimated near the stump. Growth alterations due to soil constraints were accurately reproduced using the calibrated soil module.

Conclusions Root architecture of large-size and long-living plants can be appropriately modelled with the modified RootTyp model. We suppose that diameter underestimation was due to selective acclimation to the prevailing wind, a facet not taken into account at this calibration step.

Keywords Root architectural model · Model calibration · *Pinus pinaster* · Root-soil interactions · Model evaluation · Tree anchorage

Abbreviations

A_{trans}	Transition Age (RootTyp parameter see Table 3)
BA	Branching Angle (RootTyp parameter, see Table 3)
CSA	Cross-Sectional Area of root segment
D	Root apical Diameter (RootTyp parameter, see Table 3)
GR_{init}	Initial Growth Rate (RootTyp parameter, see Table 3)
Hh	Horizontal roots on horizontal roots (Name of root type)

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Hstump	Horizontal roots branched on the stump (Name of root type)
Hvh	Horizontal roots branching from vertical or horizontal roots (Name of root type)
IBD	Inter-Branch Distance (RootTyp parameter, see Table 3)
IT	Indicator of Tropism type (RootTyp parameter, see Table 3)
L_{max}	Asymptotic length (RootTyp parameter, see Table 3)
MBIAS	Mean of BIAS (model evaluation criteria, see Table 2)
MEF	Model Efficiency (model evaluation criteria, see Table 2)
PR_{typ}	Proportion of Root types (RootTyp parameter, see Table 3)
P_{trans}	Transition Probability (RootTyp parameter, see Table 3)
RG	Radial Growth parameter (RootTyp parameter, see Table 3)
RI	Tapering-by-branching coefficient
RMSE	Root Mean Squared Error (model evaluation criteria, see Table 2)
RMSEn	normalised Root Mean Squared Error (model evaluation criteria, see Table 2)
SoilBranch	Alteration of root inter-branch distance (IBD) by the soil (Soil module parameter, see Table 4)
SoilCont	Alteration of growth direction by the soil (Soil module parameter, see Table 4)
SoilGrowth	Alteration of root elongation by the soil (Soil module parameter, see Table 4)
SoilInt	Intensity of alteration of growth direction by the soil (Soil module parameter, see Table 4)
SRL	Specific Root Length, ratio of root length to root volume
S_{trans}	Transition direction (RootTyp parameter, see Table 3)
T	Tropism intensity (RootTyp parameter, see Table 3)
TN	Time of Necrosis (RootTyp parameter, see Table 3)
Vh	Vertical roots on horizontal roots (Name of root type)
Vv	Vertical roots on vertical roots (Name of root type)

Introduction

Anchorage is one of the primary functions of the root system to guarantee plant survival. This is particularly true for forest trees whose height and long life time make them particularly vulnerable to strong winds: storms cause more than 50% of primary damage to European forests (Schelhaas 2008). Understanding the processes that underlie tree anchorage is a prime target for controlling tree stability and decreasing wind-related risks in forests. The causal factors associated with tree anchorage change as trees grow. For instance, anchorage failure in juvenile trees results in toppling (stem leaning), while in mature forests strong winds can cause damage in the form of uprooting (Coutts 1983b; Moore et al. 2008).

In trees, anchorage biomechanics is mainly governed by the 3D architecture of the root system: the mechanical resistance of the root system in the soil mostly depends on coarse root geometry and spatial distribution (e.g. the circular distribution of shallow roots as a function of the slope or the wind direction; Coutts et al. 1999; Danjon et al. 2005; Nicoll et al. 1995). The mechanical strength of the root wood and the precise characteristics of the surrounding soil are thought to be of secondary importance (Ennos 1993; Yang et al. 2018). Coarse roots can be classified as a function of their key roles in tree stability, e.g. stump, taproot, shallow and sinker roots (Coutts 1983b, 1986; Coutts et al. 1999; Crook and Ennos 1996; Danjon et al. 2005; Ennos 1993; Ray and Nicoll 1998; Yang et al. 2017). For each root category, the main factors that determine the biomechanics of the root systems are root number, root diameter, root radial distance, and root volume (Danquechin Dorval et al. 2016; Nicoll and Ray 1996).

The development of the root system architecture is firstly driven by the genetic potential of the species (Lynch 2007), but it is also plastic to allow plants to respond to their environment (Malamy 2005). The contribution of the different types of coarse roots to anchorage changes as the root system develops. In young taprooted trees species growing in deeper soils, anchorage is mainly determined by the taproot (e.g. Crook and Ennos 1996; Danquechin Dorval et al. 2016; Mason 1985; Toral et al. 2011). In older, larger specimens, anchorage is achieved by the rigid proximal part of the root system resulting in a large root-soil plate and windward roots guying the plate (Coutts 1983a, b; Danjon et al. 2005). In *Pinus* and *Picea,* this plate is composed

of the stump, the thick proximal part of shallow roots (named Zone of Rapid Taper - ZRT; Wilson 1975), the secondary sinker roots, deep roots, and the large mass of trapped soil (Coutts 1983b; Ennos 1993; Danjon et al. 2005). But the development of coarse roots also depends on the environmental factors that promote or hinder root development, i.e. soil fertility, soil geometry, a cemented horizon, or waterlogging (Tobin et al. 2007). This adaptive growth is thought to produce the high variability of the root architecture observed in situ (Danquechin Dorval et al. 2016; Horton 1958; Ray and Nicoll 1998). But soil properties are not the only environmental factors that modify root development. Root growth plasticity to the prevailing wind can occur in the form of a greater root number or a greater diameter (i.e. secondary thickening) or length to enhance tree anchorage (Danjon et al. 2005; Stokes et al. 1995; Tamasi et al. 2005). Bending resistance may also be strengthened by the formation of oval root cross-sections (Coutts et al. 1999; Nicoll and Ray 1996). In addition, adaptive responses of roots are also influenced by the soil surface geometry and roughness ensuing from soil tillage (Nicoll et al. 2006).

Different approaches have been proposed to model spatial root development; they provide a conceptual framework for root growth (see review papers by Dunbabin et al. 2013; Tobin et al. 2007). Most of these models are devoted to simulating water and nutrient uptake dynamics (e.g. Dunbabin et al. 2011; Leitner et al. 2010). Fractal-branching models are static (e.g. Van Noordwijk and Purnomosidhi 1995); they are devoted to rebuilding the root system structure from root proximal diameter. Density-based models are continuous-based approaches that model meristem dynamics (Dupuy et al. 2010). However, they do not provide the 3D root system structure required for studying the anchorage function. Root architectural models explicitly simulate the architecture of root systems in 3D space, and this makes it possible to simulate temporal and spatial dynamics. They are based on morphogenetic rules of individual roots. In these models, parameters vary either by branching order (e.g. OpenSimRoot; Postma et al. 2017; parametric L-system modelling; Leitner et al. 2010) or by root type (e.g. RootTyp; Pagès et al. 2004). Root architectural models differ in their level of complexity (Dunbabin et al. 2013), hence by the number of parameters. Architectural models also differ in the way they account for carbon allocation and nutrient and water uptake in link with plant development. Carbon costs and gains can be partially described in relation to a module for canopy photosynthesis, as in SPACSYS (Wu et al. 2007) or fully described through a functional-structural plant model (e.g. a review by Sievänen et al. (2000) or the PlaNet-Maize model by Lobet et al. (2014)). One of the main interests of architectural models is to simulate the impact of the soil (mechanics, heat, water or nutrient availability) on root development (e.g. Clausnitzer and Hopmans 1994; Leitner et al. 2010; Pagès et al. 2004). Despite some success (e.g. Landl et al. 2017), such a simulation still remains a challenge, in particular to discriminate endogenous from exogenous impacts on root system development.

Modelling architecture development is hindered by the scarcity of data available to parameterise and evaluate the model. Indeed, in the reference paper for the RootTyp model (Pagès et al. 2004), examples of simulations were not run from parameters measured on root systems but with parameters set to mimic 2D drawings made by Kutschera (1960). Moreover, three-dimensional models of root architecture have mainly been used to represent the establishment of annual crop species such as bean, lupine, maize, or wheat (Bingham and Wu 2011; Chen et al. 2011; Postma et al. 2014; York et al. 2015). In all these trials, complexity (root type number, maximum order, root number) and simulation duration (several months maximum) were relatively low. Only Jourdan and Rey (1997) simulated the establishment of a tree root system over a long period (oil palm up to 20 years, time step: 1 month). However, palm species have no secondary growth. Model parameters are generally set by measuring young plants in short-term experiments (Chen et al. 2011; Dunbabin et al. 2009; Landl et al. 2018; Pagès et al. 2013). But the establishment of the complete root system of trees, including the different types of roots, takes several years. Since non-invasive techniques such as ground penetrating radar are restricted to specific conditions (dry sandy soils, large shallow horizontal roots, isolated plants; Butnor et al. 2016), measuring the root architecture of long-life plants such as trees implies in situ studies with destructive measurements in a range of sizes and ages (Danjon and Reubens 2008). A given tree is generally measured only once. It is generally extremely difficult to retrieve all the roots of a tree grown in situ because roots rapidly grow vertically and horizontally to several meters distance, and they are entangled with

roots of neighbour trees and of the understorey. Therefore, model calibration has to cope with these experimental limits.

The aim of this paper is to design a first procedure to model the 3D root architecture of a large tree species (*Pinus pinaster*) with secondary growth over its whole life cycle, especially in order to study the dynamics of the long-life roots involved in the anchorage function. We adapted the RootTyp model (Pagès et al. 2004), which describes the main root properties. We did not use the newer ArchiSimple model (Pagès et al. 2014) because its conceptual framework assumes that the roots can be summarized by the size of their tip, which is a property that we were not able to measure on root systems in forest soils. We thus chose RootTyp because it is a generic model (used for many species), it manages the differentiation of roots in several root types, any root type can branch from any root type (unlike models which use branching order as root types) and finally, the radial growth is only controlled by one simple measurable coefficient by root type. The soil module in RootTyp aims at representing the effects of the soil on developmental processes in a very simplified way (all types of interactions, e.g. water, nutrient and oxygen distribution, or chemical toxicity, and also including mechanical root-soil interaction). Moreover, it is stochastic, so that it offers the possibility to generate variability of the root systems similar to in situ variability. We chose *P. pinaster* grown in sandy spodosols (characterized by a cemented horizon) as a model species because there exists a large database including 3D-digitized root systems of uprooted *P. pinaster* seeded in situ, with tree ages ranging from germination to clear-cutting age (50 years) (Bert and Danjon 2006; Danjon et al. 2005; Danquechin Dorval et al. 2016, and unpublished datasets). Trees were sampled in the Landes de Gascogne Forest, which covers 1 million ha of planted forest. This forest recently experienced heavy losses due to winter storms (Kamimura et al. 2015). The first stage of this work was to provide a coherent set of quantitative parameters of *P. pinaster* root development needed for simulations with RootTyp. Most of them were set from statistical analyses performed on the root system data gathered in an existing database (Table 1). Additionally, we estimated several growth parameters using the literature and an optimisation method. We also took into account the cemented horizon to integrate soil constraints into the RootTyp soil module. The second stage was to test the robustness of the RootTyp model with simulations along the life-time of a forest tree species. We assessed the reliability of simulations on global and specific root traits involved in anchorage at different ages.

Materials and methods

Root architectural model

Based on a comparison of three-dimensional root architectural models (Dunbabin et al. 2013), we chose RootTyp (Pagès et al. 2004) because it is easy to calibrate on several species with various structures (bean, lupin, maize, wheat; Bingham and Wu 2011; Chen et al. 2011; Couvreur et al. 2012; Pagès and Picon-Cochard 2014; Postma et al. 2014; York et al. 2015). Furthermore, this model had already been used to simulate the root system development of one dicotyledonous tree species at a young stage (*Quercus petraea* - Collet et al. 2006). To obtain this ability and this genericity, RootTyp combines many processes observed during root system growth (axial and radial growth, tropism, sequential branching, reiteration, transition, decay, and abscission) including radial growth with the pipe model, which is an essential feature for aged tree root systems.

As the name of the model suggests, the first step to adapt it was to determine a limited number of root types for *P. pinaster*. Seven types were described with common morphogenetic origins regardless of the branching order, which is an advantage to describe the *P. pinaster* root system (Danjon et al. 2005). Each root type has a homogeneous developmental behaviour with its own characteristic set of 12 parameters with straightforward biological significance (see below - Pagès et al. 2004 and Table 3).

Root system database

We used a chronosequence of *P. pinaster* root systems ranging from 1 to 50-years-old to calibrate and evaluate the model (Table 1). Nezer13 and Bilos50 were part of datasets used to assess the relationships between root architecture and anchorage (see Danquechin Dorval et al. (2016) and Danjon et al. (2005)). SA1, SA2 and SA3 are unpublished data sets from an experiment devoted to the comparison of planted and seeded trees over 8 years. These 5 datasets (SA1, SA2, SA3, Nezer13 and Bilos50) were obtained from trees from seeded stands

Table 1 Mean characteristics ± standard deviation of trees and their associated root systems gathered in the database. White part of the table: root systems used for calibration; italic part of the table: root systems used for model evaluation

Trees				Excavated root systems					
Tree age (years)	Dataset name	Number of trees	Stem length (m)	Collar diameter (cm)	Diameter threshold (cm)	Coarse root number	Maximum depth (cm)	Maximum radial distance (cm)	Total root volume (cm^3)
<1	Scan [c]	7	0.08 (±0.04)	0.02 (±0.00)	0.00	16.0 (±5.9)	−31.5 (±8.9)	23.0 (±11.0)	NA
1	SA1 [c]	23	0.10 (±0.00)	0.64 (±0.19)	0.01	21.0 (±6.9)	−23.1 (±5.9)	34.9 (±10.5)	4.1 (±2.6)
2[i]	SA2 [c]	13	0.77 (±0.20)	1.38 (±0.34)	0.02	23.3 (±11.9)	−47.2 (±23.3)	181.4 (±64.9)	49.5 (±44.3)
3	SA3 [c]	32	1.03 (±0.27)	1.61 (±0.64)	0.10	37.2 (±16.5)	−40.4 (±10.8)	85.9 (±38.3)	80.8 (±90)
13	Nezer13 [a]	10	9.80 (±0.80)	21.10 (±1.80)	0.50	153.5 (±48.7)	−90.6 (±11.3)	323.6 (±83.2)	28.2e 03 (±6.6e 03)
50	Bilos50 [b]	7	20.34 (±1.15)	44.90 (±8.28)	1.00	437.7 (±113.7)	−104.2 (±9.1)	298.9 (±59.3)	274.1e 03 (±117.3e 03)
1	*SA1* [c]	*6*	*0.10 (±0.00)*	*0.67 (±0.17)*	*0.01*	*17.7 (±4.8)*	*−28.9 (±7.6)*	*34.7 (±7.3)*	*4.0 (±1.4)*
3	*SA3* [c]	*5*	*1.40 (±0.43)*	*2.49 (±1.02)*	*0.10*	*49.8 (±15.0)*	*−53.1 (±13.8)*	*127.5 (±47.9*	*190.6 (±173.2)*
13	*Nezer13* [a]	*5*	*9.40 (±0.60)*	*19.90 (±1.50)*	*0.50*	*164.6 (±95.4)*	*−88.7 (±7)*	*300.9 (±30)*	*24.1e 03 (±4.8e 03)*
50	*Bilos50* [b]	*6*	*20.98 (±1.30)*	*53.02 (±3.83)*	*1.00*	*562.0 (±40.7)*	*−108.3 (±13. 5)*	*333.5 (±46.8)*	*369.9e 03 (±120.9e 03)*

[a] Root systems used in Danquechin Dorval et al. 2016

[b] Root systems used in Bert and Danjon 2006 and Danjon et al. 2005

[c] Unpublished datasets

[i] Partial root systems measured in situ

and from the same local provenance. They grew in a sandy spodosol, with a water table rising to around 50 cm depth in winter and going down to 1.5 m depth in late summer (Cottinet 1974). This soil is characterized by an organic horizon at 0–20 cm depth followed by an eluviated horizon and a discontinuous cemented spodic horizon at 40–100 cm depth. The prevailing winds come from the west. Except root systems from the scan dataset, all the other 107 root systems were excavated in winter, and the architecture was measured using a 3D digitizer according to Danjon and Reubens (2008). A root system was measured as a set of root axes subdivided into segments. The spatial coordinates and topology of all the root segments were recorded simultaneously with the root diameters in a multitree graph (MTG) format file (Godin and Caraglio 1998). The distal end of each root segment was defined arbitrarily at each lateral branch or modification in root direction, or when a rapid decrease in diameter occurred (Fig. 1). The average segment lengths were 5, 7, 10, 12, and 20 cm for the SA1, SA2, SA3, Nezer13, and Bilos50 root systems, respectively. According to the dataset age, roots with a proximal diameter smaller than a threshold limit were not measured (Table 1), but the number of these roots was recorded. The amount of roots lost during uprooting was low in the proximal part of the root system, but large in the peripheral part (Danjon et al. 2013). Yet, we succeeded in measuring several shallow roots down to the tip in SA2 (Table 1). That's why, supplementary observations on the early stage (Scan dataset; <1 year) were made with seedlings transplanted into tubes (50 cm tall, 15 cm internal diameter) filled with sandy soil. The tubes were placed vertically in a greenhouse, and root systems were pulled out, washed, and scanned 1 month after germination in order to measure roots to their tip.

We used 92 root systems for calibration and 22 root systems for evaluation (Table 1). Due to secondary growth, it was difficult to retrieve the initial branching point and angle in Bilos50, which also showed large soil constraint effects on root system development. The Bilos50 root systems were therefore not used for calibrating and evaluating the geometric parameters.

Parameters and their estimation

Most of the parameters for the different root types were included in the model as means and standard deviations computed from the database (hypothesis of normal distribution) (Table 3 – from 2D or 3D digitised root systems). But depending on data availability, different methods were used to parametrise the remaining parameters: use of literature descriptions, non-linear fit of available data, and an optimisation method.

Two parameters were calibrated using the differential evolution optimisation method (Storn and Price 1997) in the "DEoptim" R package (Ardia et al. 2016; Mullen et al. 2011). We determined the initial ranges of these two parameters from literature about conifer species (Online Resources 1a and 1b). The standard deviations of all parameters were fixed to zero to avoid replications for each combination. An initial generation of 60 root systems (the size of the population was at least ten times the length of the parameter vector) was created using

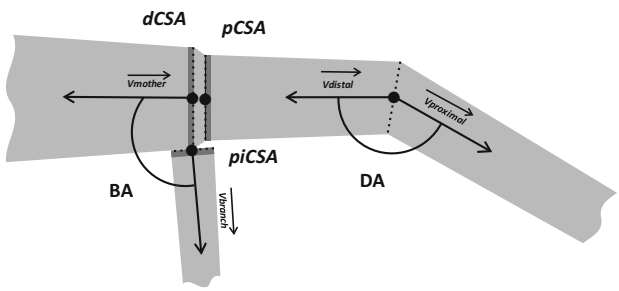


Fig. 1 Diagram of connected unit segments along a digitised root axis. Each measured segment is separated by a dashed line. The root base diameter and the distal diameter of each segment were measured to compute CSA's (cross-sectional areas). dCSA: distal CSA of the mother root segment; pCSA: proximal CSA of the root segment following the branching point; piCSA: proximal CSA of the branch root segment; Vmother: vector of the mother root direction; Vbranch: vector of the daughter root direction; Vdistal: vector of the distal root segment; Vproximal: vector of the proximal root segment. Branching (BA) and deviation angles (DA) were calculated from the above-mentioned vectors

random values within the parameter ranges. We determined the best parameter set for the model output among all the root systems of one generation by selecting the lowest root mean squared error (RMSE; Table 2) among model outputs and a selection of field-measured characteristics (longitudinal diameter distributions of first-order roots, shallow roots, and vertical roots branching in the first 1.5 m of shallow roots) at three stages (one, three, and thirteen years old). This selection and estimation of the best parameter set was run over 50 generations.

These specific methods are presented per parameter or group of parameters.

Axial growth, elongation

In the RootTyp model, a simulated root axis is a set of nodes representing the successive positions of apices at each time step. The axis can branch at any point on the corresponding segment between two nodes. The roots are assumed to elongate according to the following negative exponential function:

$$L = L_{max}\left(1-e^{\frac{-GR_{init}\,t}{L_{max}}}\right) \quad (1)$$

where L is the total length of the root axis at time t since emergence, L_{max} the asymptotic root length parameter, and GR_{init} the initial growth rate parameter.

For root types where the length to tips were available at two different stages (2 and 5 years), the asymptotic root length parameter (L_{max}) and the initial growth rate parameter (GR_{init}) were estimated using a non-linear least squares approach ("nls" function in R). For the other cases, $L_{max\ and}$ GR_{init} came from detailed descriptions given by Burbidge (1936) and from other literature sources (Online Resources 1a and 1b). For root types for which neither measurements nor literature descriptions were available, L_{max} and GR_{init} were calibrated using the differential evolution optimisation method (Storn and Price 1997, see above).

Growth variation (growth rate and root length) between branching orders has been observed by several authors (Atger and Edelin 1994; Pagès 1993). We assessed this variation of the mean length of the first four branching orders in the horizontal roots of the Bilos50 root systems.

Axial growth, direction

The growth direction of each root tip is influenced by both root tropism (IT) and tropism intensity (T). Hence, at each time step, the new growth direction results from a vectorial sum representing three influences: (i) previous direction, (ii) tropism, and (iii) tropism intensity. Each root type can exhibit positive gravitropism (a trend to keep a downward direction), negative gravitropism (a trend to grow upwards), plagiotropism (a trend to grow horizontally) or exotropism (a trend to grow toward the initial direction).

Tropism intensity was measured as the ratio between the difference in depth between the root base and the

Table 2 Criteria used to evaluate model performance

Performance criterion	Abbreviation	Formula	Ideal
Root mean squared error	RMSE	$\sqrt{\frac{\sum_{i=1}^{n}(est_i-obs_i)2}{n}}$	0
Mean of bias	MBIAS	$\frac{\sum_{i=1}^{n}(est_i-obs_i)}{n}$	0
Normalised root mean squared error	RMSEn	$\frac{RMSE}{(Max_{obs}-Min_{obs})}\times 100$	0
Model efficiency	MEF	$1-\frac{\sum_{i=1}^{n}(est_i-obs_i)^2}{\sum_{i=1}^{n}(obs_i-obs)^2}$	1
Linear regression	intercept, slope and R^2	obs_i = intercept + slope × est_i	intercept = 0, slope = 1 and R^2 = 1

R^2 = Coefficient of determination adjusted to the fitted line

est_i = ith estimated value

obs_i = ith observed value

n = number of observations

root end on the one hand, and root length on the other hand:

$$T = \frac{Z_{end} - Z_{base}}{L} \quad (2)$$

T was estimated as the mean tropism for each root type in the Nezer13 and Bilos50 root systems only, because most of their roots were rigid. The roots for which T could not be assessed accurately were excluded:

- small roots (proximal diameter < 1.0 cm), which are not rigid enough
- broken roots (<50 cm length)
- deep roots (originating at a depth >66% of maximum depth, Danquechin Dorval et al. 2016) because their growth could be heavily influenced by soil constraints.

The deviation angle defines the capacity of a root to grow in the same direction or to deviate in response to soil heterogeneity. We defined it with two vectors of 3 cm length on both sides of each segment end (Fig. 1). It was calculated as:

$$DA = cos^{-1}\left(\frac{\overrightarrow{Vdistal}.\overrightarrow{Vproximal}}{\left\|\overrightarrow{Vdistal}\right\|.\left\|\overrightarrow{Vproximal}\right\|}\right) \quad (3)$$

Branching

The branching process is acropetal, i.e. the emergence of new lateral branches follows the tip of the axis (Pagès and Serra 1994). Root branches appear at a given distance from one another, called the inter-branch distance (IBD). We calculated the IBD as the mean length separating a root base from its two surrounding neighbour branches. We estimated it for each root type.

Several authors mentioned that IBD increases when the length from the base of roots increases (Pagès and Pellerin 1994; Landl et al. 2018). We assessed this potential variation in the different root types in the Nezer13 root systems. We also examined the variation of the proportion of daughter roots along the roots.

We computed the branching angle (BA) using a vector method. We defined it for all the roots from two vectors of 10 cm length to describe both the mother root direction before the branching point (Vmother) and the basal daughter root direction (Vbranch) (Fig. 1). We computed it as:

$$BA = cos^{-1}\left(\frac{\overrightarrow{Vmother}.\overrightarrow{Vbranch}}{\left\|\overrightarrow{Vmother}\right\|.\left\|\overrightarrow{Vbranch}\right\|}\right) \quad (4)$$

We estimated BA for each root type using the subset of axes used for tropism computations.

Apical diameter and radial growth

An apical diameter (D) was given to each root when it was initiated. The apical diameter *per* root type was measured on young scanned seedlings (Table 3). Radial growth was only controlled from the apices and a tapering-by-branching coefficient (RI; Pagès et al. 2004; Soethe et al. 2007). Thus, the branching process and the apical diameter of descending roots had a major influence on the diameter of a root segment. We did not include tapering between branches in the model. RI was computed *per* root type in the Nezer13 root systems as:

$$RI = (pCSA + \sum piCSA)/dCSA \quad (5)$$

where dCSA is the distal cross-sectional area of the segment (CSA; (diameter^2 π)/4) before the branching point, pCSA is the proximal CSA of the segment after the branching point, $\sum$piCSA is the sum of CSAs of branches (measured 2 cm from the root base to ignore basal swelling; Fig. 1).

Transition

Transition is a discrete and abrupt modification of the growth pattern that occurs during root growth, e.g. a horizontal root becoming a vertical root (Eis 1974; Fayle 1975). It has been reported for several features like growth direction, branching, the proportion of lateral roots, or axial and radial growth (Arbez 1971; Heyward 1933; Pagès et al. 2004). Therefore, transition is characterised by a change in the root tip state leading to a shift from one root type to another.

In order to model this process, RootTyp considers three parameters: (i) a threshold age after which transition may occur (A_{trans}), (ii) the transition direction

Table 3 Root types and their model parameters estimated for the different processes (font types refer to estimation methods, normal = means of 2D or 3D digitised root systems, *italic* = optimisation method, and **bold** = literature). Parameters for the unmodified model or cases without any implementation of new processes are underlined

Model parameter	Stump/Taproot	Hstump	Hvh	Hh	Vv	Vh	Small	Data or method
Asymptotic length (m) – L_{max}	**3.6 ± 0.8/3.6 ± 0.8**	17.2 ± 4.1	*11. ± 2.8*	*8.8 ± 2.2*	**3.6 ± 0.8**	**3.6 ± 0.8**	**0.1 ± 0.0**	3D Digit/ *Optimization*/ **Literature**
Initial growth rate (cm/step) – GR_{init}	**8.2 ± 1.5/8.2 ± 1.5**	7.3 ± 1.2	*1.9 ± 0.5*	*3.7 ± 0.9*	*2.6 ± 0.7*	*1.5 ± 0.4*	**1.0 ± 0.0**	3D Digit/ *Optimization*/ **Literature**
Radial growth parameter – RG	0.9/0.9	0.9	0.9	0.9	0.9	0. 9	0.9	3D Digit
Root apical diameter (cm) – D	0.12/0.12	0.10	0.10	0.10	0.10	0.10	**0.04**	2D Scan/**Literature**
Inter-branch distance (cm) – IBD [Mean ± SD or Intercept-slope]	1.1 ± 0.2/ 4.2 ± 1.3	3.6–0.03 | 6.0 ± 2.0	5.2–0.03 | 7.9 ± 3.3	1.5–0.07 | 3.3 ± 1.7	3.9 ± 1.3	3.3 ± 1.6	–	3D Digit
Branching angle (degree) – BA	–/–	112.2 ± 15.7	101.4 ± 15.9	145.7 ± 14.8	145.3 ± 13.2	105.9 ± 13.5	**101.4 ± 15.9**	3D Digit/**Literature**
Indicator of Tropism type – IT	Gravi (+)/Gravi (+)	Plagio	Plagio	Plagio	Gravi (+)	Gravi (+)	**Exo**	3D Digit/**Literature**
Tropism intensity – T	0.86 ± 0.10/0.86 ± 0.10	0.11 ± 0.11	0.13 ± 0.16	0.1 ± 0.14	0.83 ± 0.15	0.89 ± 0.12	–	3D Digit
Transition age – A_{trans}	3.0/–	–	–	–	–	–	–	3D Digit
Transition direction – S_{trans}	+1/–	–	–	–	–	–	–	3D Digit
Transition probability – P_{trans}	0.9/–	–	–	–	–	–	–	3D Digit
Necrosis time (step) – TN	∞/∞	∞	∞	∞	∞	∞	∞	
Root type proportion (%) – PR_{typ}								3D Digit
Hstump	0.29 / 0.05	0.00	0.00	0.00	0.00	0.00	–	
Hvh	0.70 / 0.54	0.23*(1-Eq. 1) | 0.10	0.26*(1-Eq. 2) | 0.08	0.01	0.19	0.19	–	
Hh	0.00 / 0.00	0.22*(1-Eq. 1) | 0.09	0.39*(1-Eq. 2) | 0.12	0.04	0.00	0.00	–	
Vv	0.01 / 0.36	0.00	0.00	0.00	0.10	0.08	–	
Vh	0.00 / 0.00	0.55*(1-Eq. 1) | 0.24	0.35*(1-Eq. 2) | 0.10	0.07	0.00	0.00	–	
Small roots	0.00 / 0.05	Eq. 1 | 0.57	Eq. 2 | 0.70	0.88	0.71	0.73	–	

(S_{trans}; value −1 or + 1), and (iii) a transition probability after the threshold age (P_{trans}).

Soil constraint identification and modelling

To assess the alterations of the root system development by the cemented soil horizon, we analysed four variables: root gravitropism, root deviation, specific root length (SRL - ratio of root length to root volume), and root number observed on the seven Bilos50 trees that grew on this specific horizon. These variables were scaled within the 0–1 range and averaged in each 5% of relative soil depth layer:

$$V_{scaled} = \frac{(V - V_{Min})}{(V_{Max} - V_{Min})} \quad (6)$$

where V_{Min} is the minimum value and V_{Max} is the maximum value of the variable for each tree. For root gravitropism, the score 1 means plagiotrope direction. The relative depth is defined as a proportion of the maximum depth in each root system of the Bilos50 dataset (Table 1).

In the RootTyp soil module, four parameters (SoilGrowth, SoilBranch, SoilCont, and SoilInt), which can be modified per soil layer, can alter root growth and root type properties. These 4 soil constraint model parameters were derived from the previous observed alterations.

Elongation (L) can be altered by the SoilGrowth coefficient (SoilGrowth). Growth direction (IT and T) can be altered by SoilCont (0 = no soil constraint, +1 = constraint for horizontal growth, −1 = constraint for downward growth) and by SoilInt (0 = low, 1 = high), i.e. the intensity of alteration. When both SoilCont and SoilInt are set to 1, all the roots grow in the horizontal direction, as observed in our spodosols when the roots reach a cemented horizon (Augusto et al. 2010). The inter-branch distance (IBD) can be altered by the SoilBranch coefficient; root proliferation is obtained by setting the SoilBranch coefficient close to 0, which decreases IBD.

For model simulation, the soil was discretised in 20 layers down to a maximum depth of 1 m.

Root system simulations and evaluations

The outputs of the model were root system representations from which variables involved in anchorage (diameter, volume, length, number of roots…) were computed and compared with observed root systems with the same collar diameter. The root system development was stopped when the number of steps or collar diameter reached the desired age or diameter.

We assessed the model performance using three types of analyses that compared 15 simulated root systems for each age with the observed root systems kept for evaluation, as follows (Table 1):

(1) We assessed the reliability of the simulations of the longitudinal distribution of mean diameters on the root types most important for anchorage. We computed the root mean squared error (RMSE; Table 2) and the mean of bias (MBIAS; Table 2) to assess the average model accuracy with the same units as the variable of interest, according to Hanson et al. (2004), Legates and McCabe (1999), or Moriasi et al. (2007). RMSE quantifies the average difference between estimates and measurements. An RMSE close to zero indicates good model performance. MBIAS identifies under- or overestimated model predictions. The optimal value of MBIAS is zero, with low-magnitude values indicating accurate model simulation. Positive values indicate model underestimation bias, and negative values indicate model overestimation bias. We computed the normalised root mean squared error (RMSEn; Table 2) to provide a single index of comparison among variables of different dimensions. RMSEn measures the relative difference between simulated and observed data. The simulation is considered as excellent if RMSEn is less than 10%, good in the 10% - 20% range, fair in the 20% - 30% range, and poor if it is greater than 30% (Sorecha et al. 2017). We used modelling efficiency (MEF; Table 2) to describe how well the plot of observed data versus simulated data fitted with the identity function (1:1 line). This statistic provides a simple index of performance on a relative scale. MEF lies in the] − ∞ and 1.0] range, with MEF = 1 the optimal value. Values between 0 and 1 are generally viewed as acceptable levels of performance, whereas values

below zero indicate unacceptable performance (Hanson et al. 2004; Legates et al. 1999). We also computed the slope, intercept, and R^2 of a linear regression fitted between all the observed and simulated diameters (slope) to provide comparisons with results from the literature (Table 2). For linear regression, we assumed a slope of 1, an intercept of 0, and a high R^2 value to represent good agreement between model predictions and measurements (Hanson et al. 2004).

(2) We performed the second type of evaluation on global variables that reflect overall simulation quality (e.g. the total root volume or proximal diameter of the taproot; Brown and Kulasiri 1996; Collet et al. 2006), and on normalised variables along the depth gradient for soil constraints. We compared observed and simulated root systems of a similar collar diameter. In order to limit the effect of roots lost during uprooting, we estimated the actual volumes with models from Danjon et al. (2013). On simulated systems, we removed the smallest roots using the following base diameter threshold ((DiaAge = 1.5 (1-exp((−0.05 Age)/1.5)))) as we did in observed systems. Finally, in order to be fully effective, we made these comparisons keeping only the proximal part of the root systems (roots located within a given radial distance (RDLt) defined as a function of tree age (DistRadAge = 100 (1-exp((−10 Age)/100))) and located between 0 and 50 cm depth.

(3) Lastly, we plotted images of observed vs. simulated root systems for visual comparison (Jourdan and Rey 1997; Vercambre et al. 2003). This last comparison evaluated the geometrical aspects and the biological realism of the model.

Results

Root typology

We identified seven categories of roots (Fig. 2a and b; Table 3) from the Nezer13 trees. The *P. pinaster* embryonic or seminal root was previously described by Burbidge (1936) and Danjon et al. (1999, 2005). This root was characterised by a positive gravitropism (0.86 ± 0.10; Table 3). As mentioned by several authors, it can generate a more densely branched zone at its proximal part (Burbidge 1936; Coutts 1987; Khuder et al. 2007; McQuilkin 1935; Pagès 1993). Consequently, according to Danjon et al. (2005), we divided it into two sections, i.e. the first 25 cm ("Stump") and the distal part ("Taproot"), with a transition process. We defined the lateral root types from their growth direction and growth capacity, as mentioned by Vercambre et al. (2003) and Collet et al. (2006). We first split them into horizontal roots (gravitropism <0.5; named H* hereafter) and vertical roots (gravitropism >0.5, named V* hereafter) according to their tropism. Both categories were further split according to their branching angle, determined partly by the tropism of their mother root. Horizontal roots on horizontal roots (Hh) and vertical roots on vertical roots (Vv) had a branching angle greater than 125°. Vertical roots on horizontal roots (Vh) had a branching angle lower than 125°. Finally, within horizontal roots branched on vertical roots with a branching angle lower than 125°, we separated the larger second-order roots (basal CSA greater than one tenth of the stem diameter at breast height branching from the proximal part of the Stump/Taproot (Hstump) from the smaller horizontal roots branching from vertical or horizontal roots (Hvh). This last "Hvh" type was carried by horizontal (25%; h) and vertical (75%; v) roots.

Smaller roots (<0.5 cm base diameter in the Nezer13 trees) were just counted. They were identified as particular root types using parameters from the literature (Collet et al. 2006; Vercambre et al. 2003). This root type was non-woody, lacked secondary growth, had no branching capacity, and exhibited exo-tropism (Table 3).

Parameters estimation

Small-root proportions

The taproot was characterised by a low proportion of small roots (5%). Alternatively, large small-root proportions were observed in vertical (Vv and Vh) and horizontal (Hh) roots (71%, 73%, and 88%, respectively). A specific longitudinal small-root distribution pattern was observed along the Hstump and Hvh roots. The proximal parts of these roots branched mostly into coarse roots (60% and 40%, respectively), but this proportion decreased rapidly along the first 2 m from their base where they branched almost only into small roots (0.95% and 0.97%; Fig. 3). We included this branching

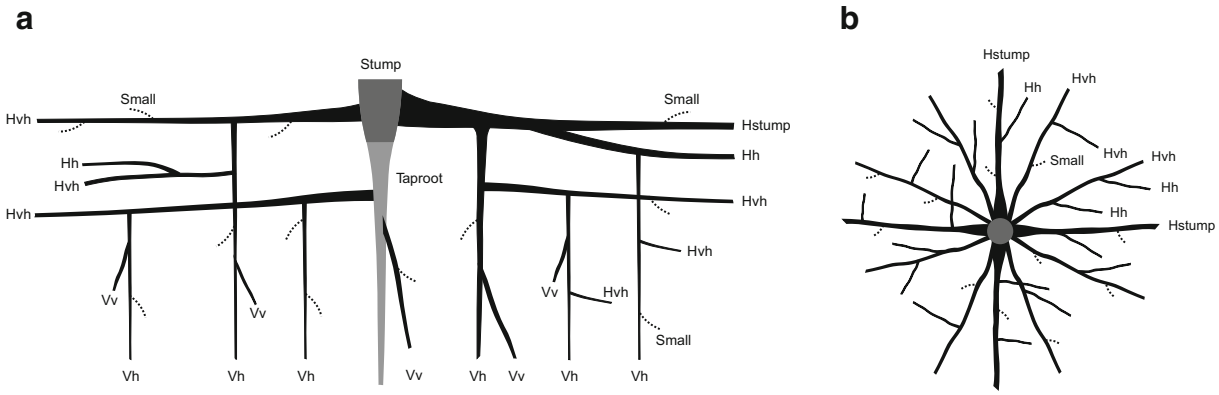


Fig. 2 Schematic representation of the maritime pine root system in (**a**) vertical and (**b**) horizontal views, with the different root types and their characteristics (First-order root: Stump/Taproot; V* vertical roots, with Vh vertical roots branched on horizontal roots and Vv vertical roots branched on vertical roots; H* Horizontal roots, with Hstump horizontal roots branched on the stump with a large CSA, Hh horizontal roots branched on a horizontal root with a branching angle larger than 125°, and Hvh all other horizontal roots)

pattern in the RootTyp model as a proportion (P) of small roots as a function of the length (L) from the root base (Table 3):

$$P_{Hvh} = 0.40\,(1-(\exp((-0.0027\,L)/0.40))) + 0.6\Big); \quad (7)$$

$$P_{Hstump} = 0.60\,(1-(\exp((-0.0045\,L)/0.60))) + 0.4\Big); \quad (8)$$

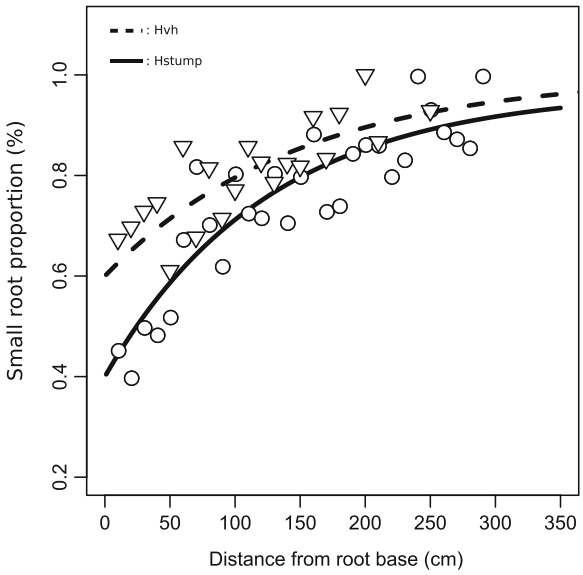


Fig. 3 Distribution of the small root proportions observed along horizontal roots (Nezer13 dataset): circles and solid line for Hstump, triangles and dotted line for Hvh (root types are illustrated in Fig. 2). The lines correspond to non-linear regression lines

Axial growth

We estimated the asymptotic length (L_{max}) parameter of the Stump/Taproot root type (3.6 ± 0.8 m) by averaging the maximum lengths of mature conifer taproots established in non-constrained soils (Online Resource 1a). As mentioned in the literature, the L_{max} parameter of Vv and Vh was supposed to be similar to the L_{max} of the Stump/Taproot (Horton 1958; McQuilkin 1935). We obtained the initial growth rate (GR_{init}) of the Stump/Taproot (0.46 ± 0.08 cm/day) from the description given by Burbidge (Burbidge 1936, Online Resource 1b) for young *P. pinaster* root systems.

For the Hstump root type, we obtained the L_{max} (17.2 ± 4.1 m) and GR_{init} (0.41 ± 0.07 cm/day) parameters using the whole length of horizontal roots measured after 2 (2.75 m) and 5 years (6 m; Danquechin Dorval 2015), adjusted with a non-linear least squares regression using Eq. 1.

As root growth models were mostly calibrated with annual crop plants, the elementary time step was generally the day (e.g. Landl et al. 2018). To simulate the growth to 50-year-old trees, we had to set a longer time step to work with acceptable computation times. This required us to use our knowledge on the transition date between the stump zone and the taproot: we set the time step to 18 days, resulting in a GR_{init} of 8.2 cm/step for the stump, so that the transition point was reached in three steps. This large GR_{init} value was similar to the mean segment size used in the observed trees. Twenty steps make 1 year, 1000 steps make 50 years.

We used the differential evolution optimisation method to estimate asymptotic length (L_{max}) and initial growth rate (GR_{init}) parameters for the Hvh and Hh root types, and GR_{init} for the Vv and Vh root types. The standard deviation of RMSEs observed over the first 5 generations was large (7.4; data not shown). This result highlights the variability of the parameter sets and root systems simulated at the beginning of the analysis. The standard deviation of RMSEs decreased in the next generations (5.8; data not shown). After 17 generations, no new parameter set was found to decrease the RMSE value. The last parameter set was supposed optimal for our model outputs and was retained (Table 3).

Branching

The inter-branch distance on the stump (IBD = 1.1 ± 0.2 cm; Fig. 4a) was low, but greater on the taproot (4.2 ± 1.3 cm; Fig. 4a). The IBD of the other vertical root types was uniform and close to the taproot mean IBD (3.9 ± 1.3 cm and 3.3 ± 1.6 cm for Vv and Vh, respectively; Fig. 4a). However, the horizontal roots were characterised by a linear increase of the distance between branches ($IBD_{Hstump} = 0.03*L + 3.6$; $IBD_{Hvh} = 0.03*L + 5.2$; $IBD_{Hh} = 0.07*L + 1.5$ Fig. 4b) toward their distal end. This IBD pattern was included in the RootTyp model (Table 4).

As mentioned in the literature, root order effects can be observed within the root system. At the adult stage, root systems are generally characterised by a limited number of orders (Atger and Edelin 1994), and the higher-order root axes often have lower asymptotic lengths and initial growth rates (Pagès and Pellerin 1994). Our Bilos50 root systems had less than 10 branching orders and showed a significant decrease in length with root order on horizontal roots (Kruskal-Wallis test; p value <2.2e-16; Fig. 5). To take this observation into account, we included a maximum number of orders and a growth coefficient *per* order in the model. We multiplied the length of a root (L) by a growth coefficient set for each order from the median length in a given branching order relative to the median length of second-order roots (60, 46, 35, 27, 21, 16, 12, and 9% for the third to tenth orders, respectively).

Radial growth and apical diameter

Observed mean tapering-by-branching coefficient (RI) was similar across all root types: it ranged from 0.84 to 1.01. Therefore, the RI parameter was set to 0.9 for all root types (Table 3; Fig. 6).

Based on measurements on seedlings, the apical diameter (D) was set to 0.12 cm for the Stump/Taproot type and to 0.10 cm for all coarse lateral roots (Table 3). The apical diameter of the small-root type (0.04 cm) was found in the literature (Collet et al. 2006; Vercambre et al. 2003).

Necrosis time

All root types were supposed perennial, so necrosis time was assumed to be infinite. Consequently, no root abscission was modelled.

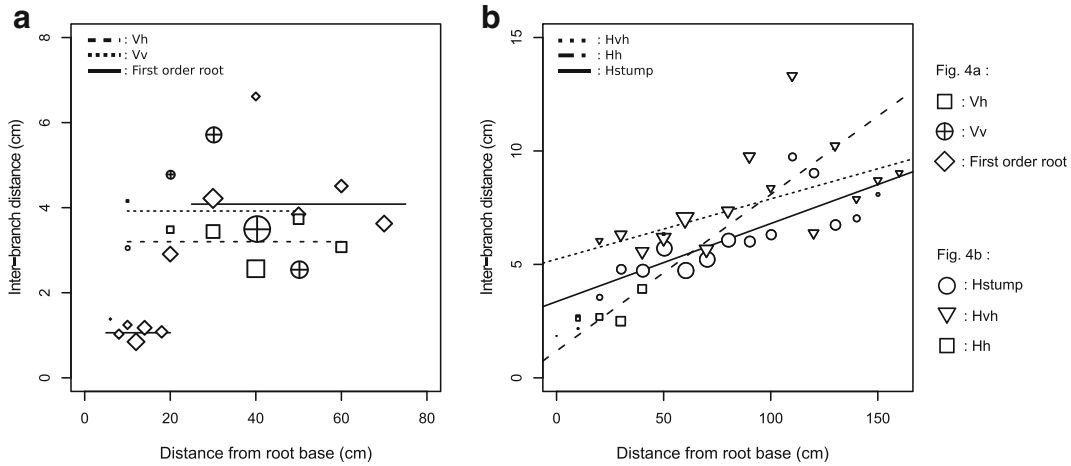


Fig. 4 Inter-branch distance observed along **a** vertical and **b** horizontal root types (root types are illustrated in Fig. 2). Symbol sizes are proportional to daughter root numbers. Mean *per* 2 or 10 cm length of the roots of ten 13-year-old trees (Nezer13 dataset). The lines correspond to the mean (vertical root types) or to the linear regression (horizontal root types)

Table 4 Mean of tree means ± standard deviations of soil constraints for (a; white part) 7 root systems from the Bilos50 dataset, and (b) 15 root systems simulated over 50 years for four horizons. Grey part of the table, soil constraint model parameters and the reference horizon for each index

Relative depth	0.0–0.3	0.3–0.7	0.7–0.9	0.9–1.0
a				
Mean scaled gravitropism	0.84 ± 0.14	0.22 ± 0.12	0.56 ± 0.17	0.75 ± 0.18
Mean scaled deviation	0.17 ± 0.19	0.24 ± 0.12	0.77 ± 0.17	0.89 ± 0.18
SoilCont value	–	−1.00 [reference]	1	1
SoilInt value	0.00 [reference]	0.09	0.58	0.79
Mean scaled SRL	0.094 ± 0.11	0.18 ± 0.12	0.54 ± 0.21	0.98 ± 0.05
SoilGrowth value	1.0 [reference]	0.91	0.51	0.02
Mean scaled root number	0.21 ± 0.17	0.29 ± 0.20	0.63 ± 0.31	0.07 ± 0.15
SoilBranch value	1.0 [reference]	0.90	0.53	0.53
b				
Mean scaled gravitropism	0.70 ± 0.02	0.54 ± 0.02	0.19 ± 0.03	0.78 ± 0.10
Mean scaled deviation	0.08 ± 0.04	0.21 ± 0.03	0.56 ± 0.03	0.97 ± 0.04
Mean scaled SRL	0.12 ± 0.08	0.29 ± 0.06	0.73 ± 0.04	0.95 ± 0.05
Mean scaled root number	0.05 ± 0.01	0.16 ± 0.03	0.27 ± 0.05	0.97 ± 0.01

Effects of soil constraints on root system architecture

In the shallow soil horizon ranging between 0.0 to 0.3 relative depths, mean scaled gravitropism was high (Table 4a; Fig. 7c). This horizon was also characterised by low root deviation, SRL and root number (Table 4a; Fig. 7a). This horizon hosts the long, thick, straight and horizontal shallow roots.

In the eluviated horizon ranging between 0.3 to 0.7 relative depth, gravitropism, root deviation, SRL, and root number showed low mean scaled values (Table 4a; Fig. 7a). In this eluviated horizon, roots were mostly vertical and are slightly less straight, thinner and more branched than in shallow soil horizon.

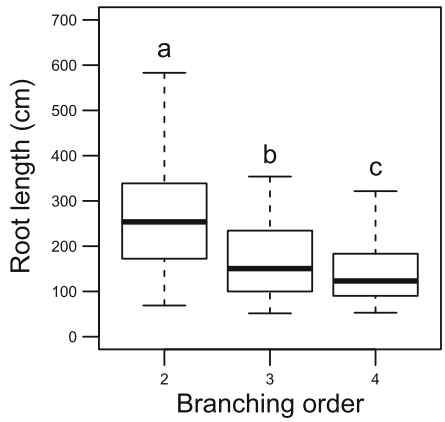


Fig. 5 Root length of horizontal roots (Hstump, Hvh and Hh; root types are illustrated in Fig. 2) observed on 50-year-old root systems (Bilos50 dataset) classified by branching order. The thick horizontal lines in the boxplots represent median values. Just below and above, the horizontal lines give the first and third quantiles. At the ends are the 5 and 95% quantiles

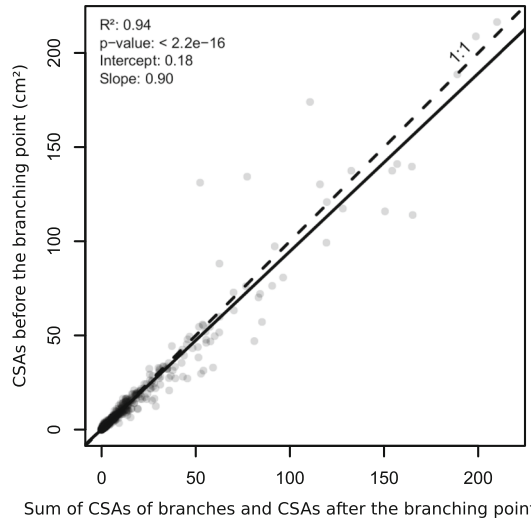


Fig. 6 Relationship between the cross-sectional area (dCSA) before the branching point on the one hand, and the sum of the CSA of branches (piCSA) and the CSA beyond the branching point (pCSA) on the other hand (Nezer13 dataset). The data of all branching points were pooled, except stump data. Solid line, model fitted on the data. Regression statistics are given

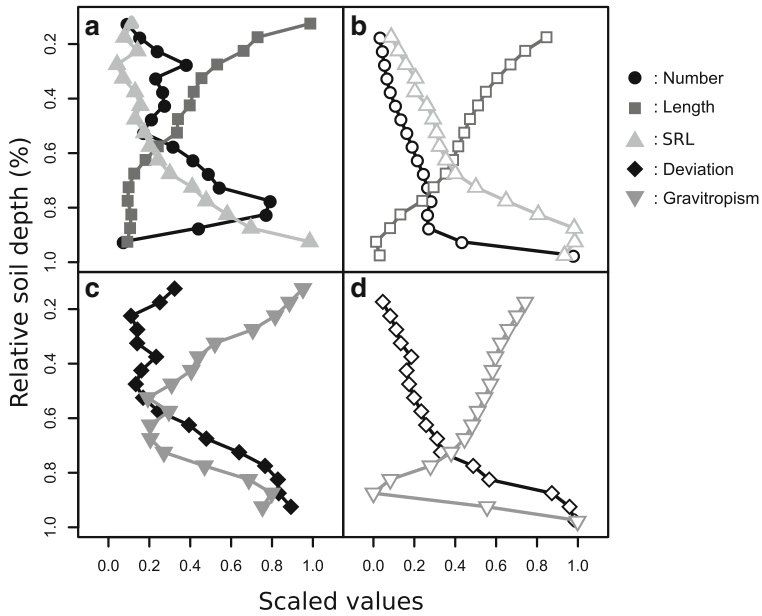


Fig. 7 Scaled mean variables (Eq. 6) *per* soil layer of 5% relative depth (0 is collar depth; 1 is the maximum depth of the root system). Gravitropism, root deviation, and specific root length were scaled and averaged on all root segments by each 5% of relative depth. Mean root length was estimated for each root and associated to the depth of its base. (**a**) and (**c**): observed means of 7 root systems from the Bilos50 dataset; (**b**) and (**d**): Means of 15 root systems simulated over 50 years with the soil constraint parameters. Observed scaled variables are solid symbols and simulated, the empty symbols

All the means of the scaled variables were highly and rapidly altered in the pre-cemented horizon (0.7 to 0.9 relative depth). In the vicinity of the cemented horizon, root moves to the horizontal direction, twist, slim down and branch more. Finally, these variables scored 0.75, 0.89, 0.98, and 0.07, respectively, in the cemented horizon (0.9 to 1.0 relative depth). The effect of the cemented horizon on roots can be observed on measured trees (Fig. 10-a4).

Based on gravitropism, we set SoilCont to −1 in the eluviated horizon and to 1 in the two deepest horizons. For the other variables, the shallow horizon was used as the reference (no soil constraint), Soilint was set to 0.09, 0.58, and 0.79 in the eluviated, pre-cemented, and cemented horizons, respectively, using the mean variation of gravitropism and deviation variables. In the same way, we set the SoilGrowth parameter to 0.91, 0.51, and 0.02, respectively, based on the mean variation of SRL. We set SoilRamif to 0.9, 0.53, and 0.53, respectively, using the variation of root numbers except for the cemented horizon where we kept the alteration of the pre-cemented horizon.

Root system simulations and evaluations

Checking of improved model performance

We added several processes observed but not initially included into the RootTyp model to simulate the root system development up to the mature stage with the best possible accuracy. We performed optimisations using this "complete" model to obtain a complete parameter set with the best calibration. To evaluate if adding the observed processes improved the ability of the model to represent *P. pinaster* root systems, we first run the unmodified model (= case1) and then run the other three cases, including the three observed processes one by one:

- A model including the reduction of growth length *per* order and the maximum branching order (= case 2)
- A model including the previous modification + an increase of the small-root proportion (= case 3)
- A model including the previous modifications + a decrease of inter-branch distance (IBD) (= case 4; "complete" model)

We assessed the similarity of the real vs. modelled root systems by generating 15 root systems aged 3 and 13 years examining statistical indices for longitudinal diameter distributions for the Stump/Taproot and shallow roots.

The parameter sets used for the different model versions are described in Table 3. The diameters of the Stump/Taproot were highly overestimated by the original model when trees were 3 years old (case 1) (RMSE = 3.22 cm, MBIAS = 2.62 cm, and MEF = −16.54;

Table 5), and were grossly over-estimated when trees were 13 years old. The 3 modified model versions gave quite similar RMSE values for the Stump/Taproot when trees were 3 years old (1.03, 0.94, and 1.18 for cases 2, 3, and 4, respectively; Table 5). For 13-year-old trees, only case 4 yielded low RMSE's and positive MEF's (2.01 and 0.58 for RMSE, and 0.90 and 0.81 for MEF for the Stump/Taproot and Hstump types, respectively; Table 5). Therefore, the "complete model" was used for further evaluation.

Model evaluation

Specific evaluation in link with tree anchorage In the Nezer13 and Bilos50 trees, the diameter of the stump decreased rapidly in the first 25 cm (Fig. 8a). The simulated diameters were in agreement with measurements at all ages (RMSEn <20% and MEF >0.5 at all stages, Fig. 8a; Table 6). However, at older stages (13 and 50 years old), the model systematically underestimated the diameter of the basal part (ca. <60 cm; Fig. 8a), with an MBIAS of −1.92 and −5.67 cm when trees were 13 and 50 years old, respectively (Table 6).

In the 13-year-old trees, the Hstump and Hvh branching from the stump (Fig. 8bc) showed moderate tapering at their base, and the diameters were correctly simulated, with RMSE's of 0.68 and 0.14 cm and MBIAS's of 0.32 and 0.05 cm, respectively (Fig. 8bc; Table 6). However, in the Bilos50 trees, tapering was very rapid, with large basal diameters extending to small diameters beyond one meter from their base. This particular observed diameter distribution was not simulated precisely by the model (RMSEn = 21.21 and 29.10%, and MEF = 0.52 and − 0.03 for Hstump and Hvh branching from the stump, respectively, Table 6). The simulation underestimated the diameters in the basal part and overestimated them beyond 40–60 cm from the root base (Fig. 8b). The longitudinal diameter distributions of Vh branched on the first 1.5 m of both Hvh originating from the stump and Hstump were simulated correctly for 13- and 50-year-old trees, with RMSEn's of 15.49 and 11.32 and MEF's of 0.75 and 0.83, respectively (Fig. 8d; Table 6). The vertical roots branched on the basal part of Hvh, and Hstump had a basal tapering lower than the other root types.

Global evaluation When trees were 1 and 3 years old, the Stump/Taproot root volumes were correctly simulated (Table 7a). However, the Stump/Taproot root volume of 13- and 50-year-old trees was underestimated (Table 7b and c).

Conversely, the volumes of Hvh branched on the stump (Table 7b and c) and Vh branched on the Hstump and Hvh branched on the stump in 13- and 50-year-old trees (Table 7b and c) were largely overestimated.

The Hstump volumes in 13- and 50-year-old trees (Table 7b and c) were correctly simulated.

When trees were 1 and 3 years old, the total root system volumes (Table 7a) were correctly simulated, and so was the total root system volume of 50-year-old trees (Table 7c). Yet, the total root system volume of 13-year-old trees was overestimated by the model (Table 7b).

In the Nezer13 and Bilos50 root systems, variability among real root systems was very high, much higher than among modelled root systems for both the number and the diameter of coarse third-order roots branching both from Hvh branched on the stump and from Hstump (Fig. 9). The number of roots sharply decreased with the radial distance class within 2 m radial distance. In the modelled root systems; the mean basal diameter was stable or decreased slightly with radial distance. Conversely, it decreased sharply in observed trees (from

Table 5 Values of criteria for case model comparisons based on the distribution of diameters as a function of length on the stump/taproot and Hstump of 3- and 13-year-old trees. The differences between model cases are explained in the text

Year	Three-year-old				Thirteen-year-old							
Root type	Stump/Taproot				Stump/Taproot				Hstump			
Case	1	2	3	4	1	2	3	4	1	2	3	4
RMSE (cm)	3.22	1.03	0.94	1.18	–	4.12	1.71	2.01	–	8.14	1.62	0.58
MBIAS (cm)	2.62	0.99	0.91	1.13	–	2.58	−0.32	−0.69	–	8.12	1.56	0.54
MEF	−16.54	−0.78	−0.50	−1.35	–	0.58	0.93	0.90	–	−36.87	−0.50	0.81

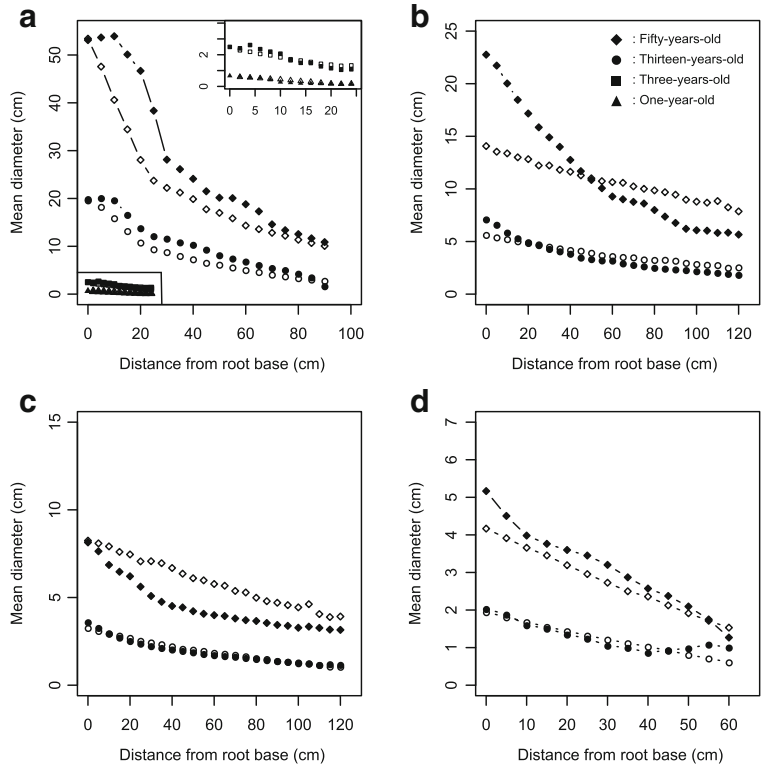


Fig. 8 Observed (solid symbols) and simulated (empty symbols) mean diameters *per* root type: **a** stump/taproot, **b** Hstump, **c** Hvh branched on the stump, and **d** Vh branched on the first 1.5 m of Hvh and Hstump branched on the stump (root types are illustrated in Fig. 2) at different ages (diamonds = fifty-year-old (6 observed and 5 simulated root systems); circles = 13-year-old (5 observed and 15 simulated root systems); triangles = 3-year-old (5 observed and 15 simulated root systems); squares = 1-year-old (6 observed and 15 simulated root systems)). The observed and simulated root systems had similar collar diameters. The inset in Fig. 8a is a zoom on one- and three-year-old trees

1.57 ± 1.2 to 0.75 ± 0.33 in Nezer13 trees, and from 6 ± 4.1 to 2.6 ± 0.8 in Bilos50 trees; Fig. 9).

Evaluation of soil constraints The soil constraint simulation correctly reflected the presence of a cemented horizon for root deviation, SRL, and root length when compared to field observations (Table 4; Fig. 7b and d). The cemented horizon depth was constant for all simulated trees (Fig. 10b2, b3, b4), conversely, the depth at which growth was altered by the cemented horizon varied substantially within each observed tree (Fig. 10a4), as only few of their roots grew deeper than 0.9 relative depth. That is why the number of roots decreased largely in the cemented horizon.

Table 6 Values of the criteria used to compare the performance of the model simulating diameter distribution *per* root type

Root type	Age (year)	RMSE (cm)	MBIAS (cm)	RMSEn (%)	MEF	Intercept	Slope	R^2
Stump/Taproot	1	0.09	0.05	17.60	0.75	0.11	0.80	0.84
Stump/Taproot	3	0.23	−0.01	14.59	0.82	0.69	0.59	0.96
Stump/Taproot	13	2.26	−1.92	12.25	0.83	−0.76	0.89	0.96
Stump/Taproot	50	7.88	−5.67	18.31	0.74	0.86	0.77	0.89
Vh[h]	13	0.18	−0.03	15.49	0.75	−0.05	1.01	0.81
Vh[h]	50	0.44	−0.34	11.33	0.83	0.42	0.76	0.97
Hstump	13	0.68	0.32	12.92	0.78	1.69	0.61	0.96
Hstump	50	3.63	−0.56	21.21	0.52	7.12	0.32	0.95
Hvh[s]	13	0.14	0.05	5.74	0.95	0.11	0.97	0.96
Hvh[s]	50	1.45	1.35	29.11	−0.03	1.90	0.88	0.87

[h] = selected Vh: Vh branched on the first 1.5 m of Hstump and of Hvh branched on the stump

[s] = selected Hvh: Hvh branched on the stump

Table 7 Comparison of root volumes between the observed and simulated root systems of (a) one-, three-, (b) thirteen-, and (c) fifty-years old pine trees for several root types and total root systems. Root volume between 0 and 50 cm depth and within a specific radial distance according to each stage. The observed and simulated root systems had similar collar diameters

a					
Collar diameter (cm)	0.67 ± 0.2		2.49 ± 1.0		
Root type	Stump/Taproot	Total root system	Stump/Taproot	Total root system	
Observations (cm^3)	2.2 ± 0.7	2.7 ± 0.9	80.1 ± 80.7	121.5 ± 121.6	
Simulations (cm^3)	2.5 ± 0.3	3.6 ± 0.4	68.8 ± 7.3	140.8 ± 16.9	
b					
Collar diameter (cm)			19.88 ± 1.5		
Root type	Stump/Taproot	Hstump	Hvh^s	Vh^h	Total root system
Observations (dm^3)	8.1 ± 1.1	4.6 ± 1.1	2.8 ± 1.6	0.7 ± 0.3	17.5 ± 4.1
Simulations (dm^3)	5.8 ± 0.8	5.2 ± 2.4	4.1 ± 0.9	2.6 ± 0.5	25.3 ± 3.3
c					
Collar diameter (cm)			53.02 ± 3.8		
Root type	Stump/Taproot	Hstump	Hvh^s	Vh^h	Total root system
Observations (dm^3)	66.5 ± 19.8	51.4 ± 25.6	9.5 ± 5.8	6.9 ± 5.0	186.4 ± 30.0
Simulations (dm^3)	37.7 ± 4.7	46.9 ± 11.7	30.0 ± 8.6	16.3 ± 1.8	195.4 ± 4.0

^{s} = selected Hvh: Hvh branched on the stump

^{h} = selected Vh: Vh branched on the first 1.5 m of Hstump

Additionally, to compare the values of the variables in the observed and modelled root systems, we removed <1-cm-base-diameter roots from the modelled root systems (threshold of measurement in the observed root systems - Table 1), i.e. most of the deep horizontal roots. However, the deep part of the secondary sinkers was not removed. In observed trees, all these roots were kept because they were thicker than this chosen threshold.

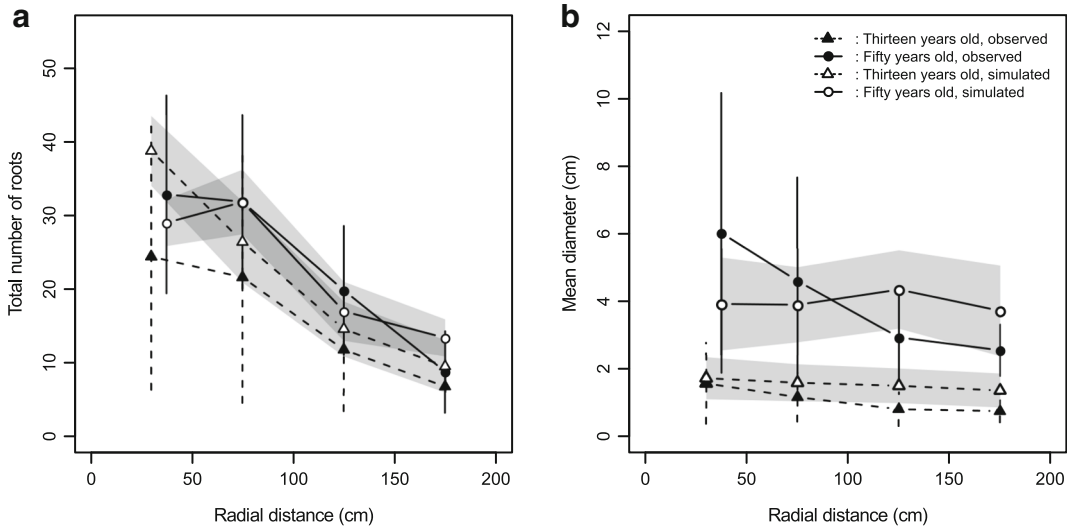


Fig. 9 Coarse third-order roots branching from Hvh branched on the stump and from Hstump in observed (6 root systems from thirteen-year-old Nezer13 pine trees and 5 root systems from fifty-year-old Bilos50 pine trees; solid symbols) and simulated (15 root systems from thirteen-year-old pine trees and 5 root systems from fifty-year-old pine trees; empty symbols) means and variability of (**a**) the total number of roots and (**b**) their basal diameter. Triangles and dashed lines: thirteen-year-old root systems. Circles and solid lines: fifty-year-old root systems. The standard deviations are represented by vertical lines for observations and envelopes for simulations. The observed and simulated root systems had similar collar diameters

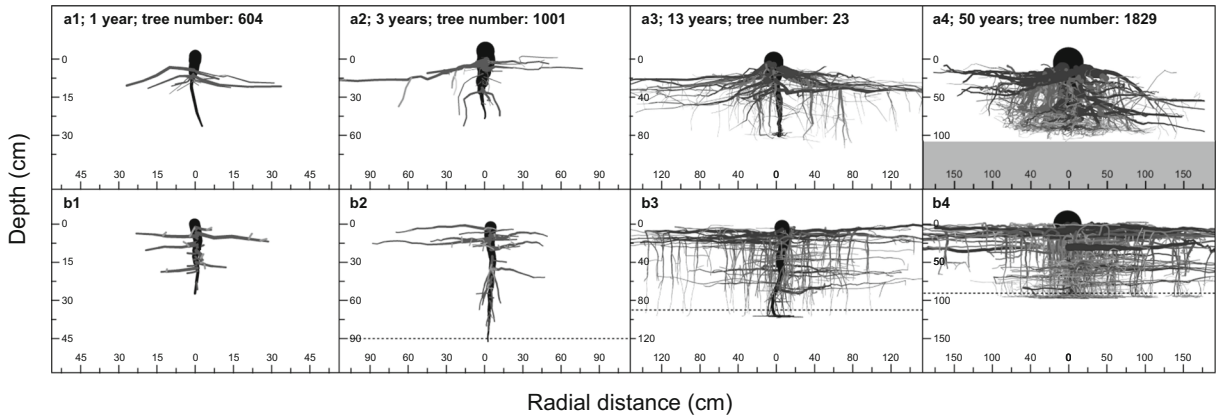


Fig. 10 Examples of observed (top, **a**) and simulated (bottom, **b**) root systems over (1) one, (2) three, (3) thirteen, and (4) fifty years. For observed tree 1829 (a4), the maximum depth is indicated by the upper limit of the grey area. For simulated trees, the upper limit of the cemented horizon is indicated by the dotted horizontal line (90 cm depth - b2, b3 and b4)

This dimensional bias explains why gravitropism was very low and the standard deviations were large in observed root systems around 0.9 relative depth. As a consequence, in the pre-cemented horizon, both the gravitropism and root number index means of modelled trees were much lower (0.19 and 0.27, respectively; Fig. 7) than in Bilos50 trees (0.56 and 0.63; Table 4 - Fig. 7).

Discussion

Model parameterisation and root system database

In developmental models based on root types, the modeller needs to set quite a lot of parameter values that are often hard to gather. Several sources of information have to be used. In herbaceous plants or young trees, root distributions generally come from partial excavations followed by root drawings, manual or digitised measurements, scans, field [mini-]rhizotrons, rhizotron boxes, or CT-scans (e.g. Collet et al. 2006; Van Dusschoten et al. 2016; Jourdan and Rey 1997; Landl et al. 2018; Leitner et al. 2010; Mairhofer et al. 2013; Mooney et al. 2012; Pagès et al. 2013; Vercambre et al. 2003). As we were mainly interested in the anchorage function of trees beyond the juvenile phase, and as we worked in heterogeneous forest soil with many understorey roots, we worked on a chronosequence of full 3D digitised proximal parts of uprooted root systems with a substantial number of measured trees. Thus we aimed to better appraise both intra- and inter-tree variability (see Fig. 6 in Danquechin Dorval et al. 2016). The database provided decent distributions for the growth direction or diameter growth variables, but was poorly adapted for gathering information on root elongation because the distal part of roots was generally lost during uprooting. We only succeeded in measuring a few shallow roots down to the tip in 2- and 5-year-old trees. We considered that gathering literature data or using an optimisation method were well adapted to set the values of a small number of parameters when modelling the root systems of larger trees, while gathering this information through measurements would have been too tedious. In situ rhizotron or rhizotron box data may provide more accurate distributions for some of the parameters, but root growth is disturbed by the device, root type discrimination is not easy, and only 2D growth can be recorded.

We set each parameter value from the root systems in the database with the best suited developmental stage. In young uprooted trees, we failed to measure the geometry of finer roots precisely, as noticed by Tamasi et al. (2005), and some root types were not yet present or too fine to be gathered or measured (Atger and Edelin 1994; Coutts and Lewis 1983; Jourdan and Rey 1997; McQuilkin 1935). It should be noted that, except for the trees grown in tubes, our database was not established for this modelling work. Finally, most of the parameter values were set from the 13-year-old trees, where measurements are more precise and diameter thresholds smaller than in the Bilos50 root systems. Moreover, at this developmental stage, the effect of root growth plasticity in response to the prevailing wind is less pronounced and thereby allows for better discrimination of intrinsic vs. environmental effects. However,

we succeeded in performing a fairly reliable simulation up to the mature stage. This was certainly due to the fact that *P. pinaster* generally do not grow adventitious roots (Zobel and Waisel 2010) or retarded branchings (Atger and Edelin 1994). Thus the proximal part of their root system is established after a few years and will then only change through secondary growth. Additionally, we only tested the reliability of simulations on the proximal part of roots, which is predominant for the anchorage function (Weber and Mattheck 2005). In species with adventitious rooting or retarded branching, some parameters describing the corresponding root types would certainly have to be set from measurements at the mature stage.

We described root growth alteration due to the soil properties variations using 4 variables varying as a function of depth, using the mature root system data. We obtained a fairly reliable simulation of this alteration by setting the 4 constraint indices in each 5% rooting depth layer. Alterations in the eluviated horizon could be appropriately modelled. However, due to intra-tree variability of cemented layer depth and strong acclimation of roots to mechanical stimuli in deep layers (Danjon et al. 2005), the model failed to precisely account for some of the growth alterations. This way of modelling the soil influence could certainly be used for other soil properties modifying root growth like a water table, rocky subsoil, increasing soil impedance, or uneven nutrient distribution. Using the soil module is thus an easy way to model shallow vs. deeper rooted root system, to explore the incidence of soil depth on anchorage.

Modelling root system development up to the adult stage

We used a long time-step (18 days) for modelling, resulting in less than 8.2-cm-long segments. We assumed that it had no influence on modelling quality and precision if it was used for studying anchorage. However, the incidence of time step duration on the unitary segment size may be a problem when studying water transport or carbon fluxes, among others. Schnepf et al. (2018) overrode this pitfall in the CRootBox model in which segment length does not depend on the time step.

In models used by Diggle (1988), Dunbabin et al. (2002), Leitner et al. (2010), or Schnepf et al. (2018), root types are defined based on the branching order. In the same way, the typology used by Vercambre et al. (2003) and Collet et al. (2006) for lateral roots also more or less encompassed the branching order. Thus, the corresponding parameters accounted for the decrease in vigour with increasing branching order. This decrease has been reported in the root systems and shoots of several species (Atger and Edelin 1994; Barthélémy and Caraglio 2007; Pagès et al. 1993). Alternatively, Pagès et al. (2013) implemented this decrease separately from the root types by attributing a growth potential to each meristem depending on the size of the mother root. We simply implemented the growth reduction observed in *P. pinaster* root systems as a growth reduction coefficient that decreased with increasing root order. This implementation seemed to be efficient and essential to simulate root system development until trees are 50 years old.

The inter-branch distance (IBD) along the Stump/Taproot decreased suddenly after 20–25 cm of growth. In *P. pinaster* the IBD of the stump in cuttings is twice the IBD observed on seedlings: the high branching rate of the stump has therefore a morphogenetic origin (Khuder et al. 2007). This particular behaviour seems to be efficiently modelled by a transition process proposed by the RootTyp model. On the other hand, IBD along horizontal roots predominantly increased with increasing distance from the root base. A similar increase was observed in several species (e.g. *Pinus resinosa* by Fayle (1975); *Pinus* species by Strong and Roi (1983); *Zea mays* by Ito et al. (2006), Pagès and Pellerin (1994), and Postma et al. (2014); *Pisum sativum* by Tricot et al. (1997), and *Triticum aestivum* by Landl et al. (2018)). Soil compaction (Pagès and Pellerin 1994), lack of oxygen (Liang et al. 1996), or lower water availability (Bao et al. 2014) may alter the branching density and can thus explain this increasing IBD. But in our case, the soil properties only marginally changed when the root grew further from the stump, therefore the increased IBD had certainly an endogenous origin. Our findings suggest that IBD along *P. pinaster* horizontal roots follows a linear trend that a transition process cannot model. This contrasts with common assumptions of 3D root architecture models in which IBD's are either set to a fixed value or drawn from a normal distribution. Taking this process into account seems to have significantly improved the model performance with *P. pinaster*.

The proportion of a given root type emitted by roots is an essential parameter included in few 3D root architecture models (e.g. Barczi et al. 2018; Schnepf et al.

2018). RootTyp is flexible about branching relations: a given root type can carry different root types unlike models using only branching orders. Our findings suggest that the proportion of small roots increased rapidly after a few decimeter growth during axial elongation. A similar trend was also observed by Fayle (1975) and Strong and La Roi (1983) in conifer species for shallow roots. One reason may be that the anchorage function requires larger roots in the proximal part of the root system only (Weber and Mattheck 2005; Danjon et al. 2005), whereas the distal part roots serve mainly for water and nutrient absorption and symbiotic interactions which are mainly ensured by small roots. This trend is poorly documented in the literature even though it seems to be an important process for tree root system modelling up to the mature stage.

Unlike already modelled smaller plants, taking into account the three above-mentioned "damping" functions turned out to be necessary to achieve reliable simulations in large trees. Like shoots in a tree crown, root vigour progressively decreased. Similarly to the formation of a cage delimiting a potential root-soil plate, damping of growth with branching order and distance from the root base is probably shared by most of the larger tree species. Damping of longitudinal growth is already a standard feature in RootTyp (negative exponential).

Modelling of root diameter up to the adult stage

As most root models are used to study water and nutrient uptake (Couvreur et al. 2012; Dunbabin et al. 2011; Leitner et al. 2010), model evaluation was generally done on variables like the distribution of root length density, the root surface, or the number of root tips (Zhao et al. 2017). Alternatively, we evaluated the model from the spatial variation of variables known to play a significant role in anchorage in the proximal part of the root system.

The role of root diameter in anchorage is well documented in the literature (e.g. Coutts et al. 1999; Danquechin Dorval et al. 2016; Yang et al. 2017). However, up to now, root modellers have not paid much attention to diameter growth (Schnepf et al. 2018). These models were mainly designed to study water and nutrient absorption and were used on annual plants or palm trees with no or poor secondary growth. Therefore, diameter growth is often included as a growth rate or a ratio varying as a function of the distance along the root (Barczi et al. 2018 [DigR]; Postma et al. 2017 [OpenSimRoot]; Schnepf et al. 2018 [CRootBox]). In both of Pagès's models (RootTyp and ArchiSimple; Pagès et al. 2004; Pagès et al. 2014), diameter growth is computed recursively only from the downstream meristem diameters and from tapering by branching (Soethe et al. 2007). Our results suggest that by handling diameter growth in that way, the diameters of the root system are generally properly modelled, provided that damping functions are used in higher orders and in the distal part of the root system. The damping functions have to be established from data within 3 m radial distance. However, as observed by Soethe et al. (2007), beyond the seedling stage, root diameter tapers between branches in the proximal part, which is often a plastic response (acclimation) to mechanical stimuli (i.e. stem swaying). In trees, the proximal parts of shallow roots branching from the stump named "zone of rapid taper" (see Fig. 2 in Coutts 1987), show mechanical reinforcement in response to wind movement. This thigmomorphogenesis process to increase investment in diameter growth takes also place at the stem base, the stump and the secondary sinker bases. It was exactly in the stump and in the proximal parts of shallow roots (Hstump and Hvh branching from the stump) that our model under-estimated diameters. However, we had no data on the root systems of trees whose stem was guyed over a long period to conclusively attribute this discrepancy to adaptive growth to mechanical stress (Niez et al. 2018).

In the same way, the diameter of third-order roots strongly decreased within 2 m radial distance in the measured trees, but it did not do so in the modelled trees because all these roots were of the same type and hence had the same parameter values and vigour whatever their radial distance. The only size decrease was due to age differences, which were low in this zone. This also explains why the Hstump and Hvh diameters were largely overestimated by the model beyond 50 cm distance from their proximal part: the growth parameters for the third-order roots were set from means computed in the central part of the root system, and therefore certainly largely overestimated. This additional variation was probably again due to acclimation of the central part of the root system, reinforcing the roots forming the vertical elements of the cage in mature trees. Finally, deep roots had a thicker basal diameter in the observed trees, certainly due to mechanical acclimation to the cemented horizon on which they rest (Danjon et al.

2005). We did not include the variations attributed to mechanical acclimation in the model at this step.

Finally, it should be noted that we compared observed and simulated trees of same sizes using the collar diameter as a size variable. However, strong acclimation takes place at the collar, therefore the size of the modelled trees used for comparison tends to have been overestimated (smaller collar diameter as compared to acclimated trees of the same age).

Another specificity of forest trees is that unlike crops or monoclonal plantations, tree size variability is generally large within a stand. We mainly parameterized the model with the NZ13 dataset composed of only the largest trees in the stand. Moreover, a large variability in root vigour is also generally observed (see Figs. 3 and 4 in Coutts et al. 1999). A better reproduction of the natural variability of root systems may thus be achieved by implementing a vigour coefficient modulating all parameters depending on tree and root vigour.

Conclusion and perspectives

Three-dimensional models of root architecture have mainly been used to represent the establishment of annual crops to study water and nutrient uptake. The establishment of the complete root system of large trees takes several decades, and an accurate modelling of diameter growth is required to yield reliable biomass values and mechanical properties of mature trees. Therefore, modelling tree root systems from seed to the mature stage also remains challenging because it requires data from all developmental stages.

Using a database gathered for one large tree species, we were able to parametrise the generic RootTyp model with seven root types to yield reliable simulations up to the mature stage, provided that several damping processes are added to the original model. Most of RootTyp parameters were set from statistical analyses performed on the root system database. But, owing to some limitations of this database, the literature and an optimisation method had to be used to estimate the other parameters. Managing diameter growth using both a unique tapering-by-branching coefficient and a meristem diameter per root type provided a simple and satisfactory way to model this important feature for anchorage all over the root system and throughout tree life. The interaction with variations in soil properties along the soil profile were also modelled satisfactorily using the soil module.

Besides the validation of global root system features, we also checked the reliability of longitudinal diameter distribution per root type in the simulations because the central part of the observed root systems was entirely measured. We found discrepancies between the data and the model for these distributions, certainly due to selective acclimation to the prevailing wind. Consequently, RootTyp seems adapted for modelling root system establishment in tree species, taking soil properties into account, provided that damping of growth is taken into account for modelling beyond the juvenile phase.

To better simulate the root system development and to include one major driver of the establishment of the root architecture of the tree root system, acclimation to the prevailing wind still remains to be implemented in the model. We propose to use simplified root system patterns derived from RootTyp simulations as an input in a biomechanical model (RootAnchor; Yang et al. 2014). This biomechanical model would provide information about the spatial distribution of the mechanical strains due to stem swaying. These strains could be compared to the distribution of observed plastic responses to the prevailing wind. In the long term, this approach should provide functions to the RootTyp model to simulate plastic responses to the prevailing wind (mainly alterations of secondary growth).

To further assess the ability of RootTyp to represent variability in *P. pinaster* root architecture, we should compare observed and simulated root systems established in the absence of soil constraints (e.g. root systems of trees grown on coastal sand dunes). This comparison should help to evaluate if endogenous and exogenous factors were correctly discriminated.

In the same way, a sensitivity analysis could be made to test which parameter distribution (e.g. asymptotic apical growth, branching capacity) has to be modified to account for the genetic variability between *P. pinaster* provenances (Danjon et al. 2009).

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